PCA-BASED ANALYSIS

March 30, 2026

Although both dictionary learning and Independent Component Analysis (ICA) presented evidence suggesting that separator and answer tokens exist in the same subspace, they do not provide clue on the proper subspace associated with the ICL tasks. To address this issue, we repeated our analysis using Principal Component Analysis (PCA). In PCA, principal components are sorted according to the magnitudes of eigen values. Further, it is notable that components with high eigen values mediate more information than ones with lower eigen values. That is, if we can identify a set of principal components with high eigen values, which are shared among different token types (query, separator and answer tokens), they could explain the subspace, by which large language models (LLMs) can use to do ICL tasks.

In this analysis, we analyzed the residual streams of all three types of tokens in ICL prompts (query, separator and answer tokens) by using PCA. Specifically, we collected residual streams from all layers and obtained 30 principal components (i.e., "n_component=3" in scikit-learn setting/terminology) for each type. That is, PCA components are independently obtained from query, separator and answer tokens. They refer to as v^Q (query), v^S (separator) and v^A (answer) below.

Fig. shows the cosine distance between 30 principle components of query (v^Q) and answer (v^A) tokens. As shown in the figure, 2-3 principal components of answer and query tokens are similar with each other. As shown in Fig. A, the first two principal components between v^Q and v^A are similar with each other (in terms of cosine distance). We note that the cosine distance between the third principal components of v^Q and v^A is high in some cases. This can be explained by that the cosine distance between $\vec{v_1}$ and $\vec{v_2}$ is maximized to be 2 when $\vec{v_1} = -\vec{v_2}$. As $\vec{v_1}$ and -$\vec{v_1}$ can model the same subspace, we further tested the $D' = min(D, 2 - D)$, where D is the cosine distance (Fig. B).

We project the residual streams of query, separator and answer tokens in the last layer to subspace spanned by two principal components (Fig.). We made three observations. First, all three token types are clustered well according to their types. Second, the query and answer tokens are weakly separated. Third, the separator tokes are better separated from other token types. These results raise the possibility that query, separators and answer tokens are mapped onto clusters in subspace of LLMs' activation space. If tokens are mapped onto the same clusters in a subspace depending on their types, ICL tasks could be solved by vector algebras in the subspace.

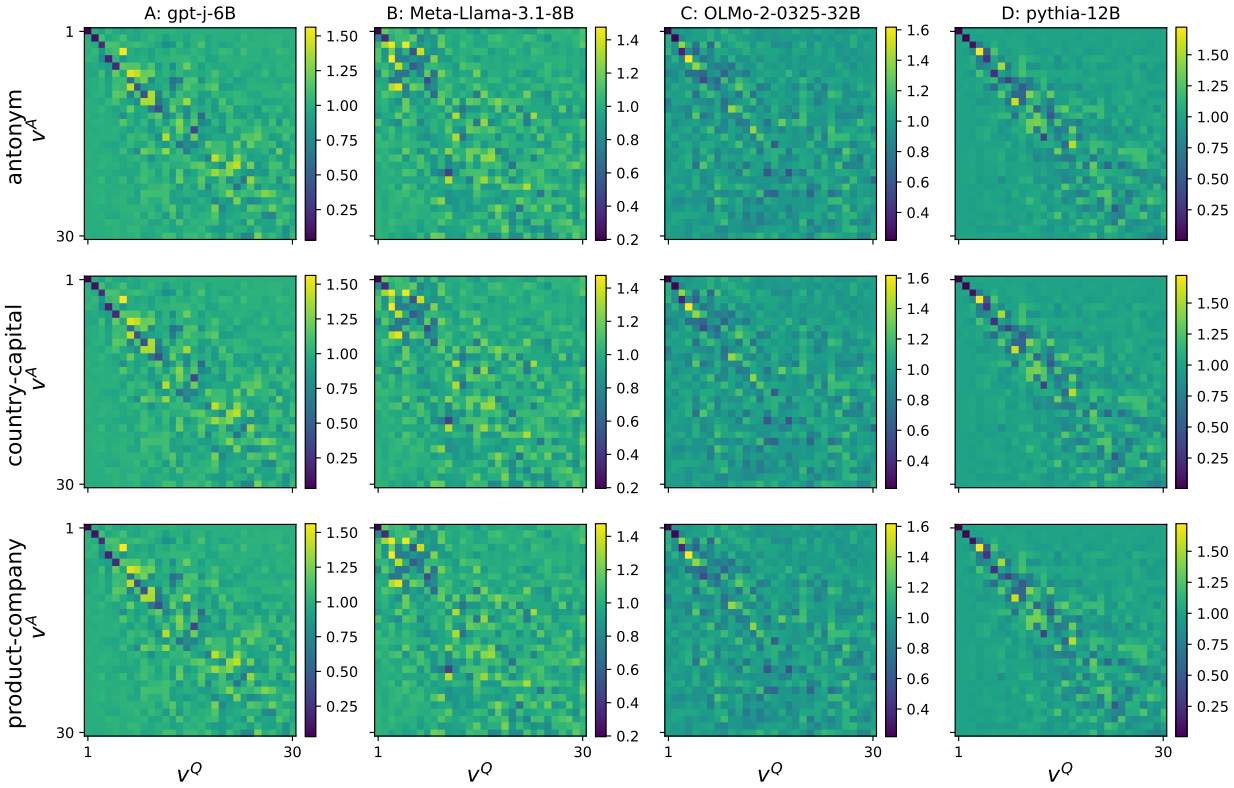

Figure 1: Cosine Distance between query and answer tokens. We computed principal components (v^As, v^Ss, v^Qs) from answer (A), separator (S) and query (Q) tokens. Panels (A)-(D) shows the cosine distance calcuated from gpt-j-6B, Meta-Llama-3.1-8B, OLMo-2-0325-32B and Pythia-12B. Top, middle and bottom rows denote 'antonym', 'country-capital' and 'product-company'.

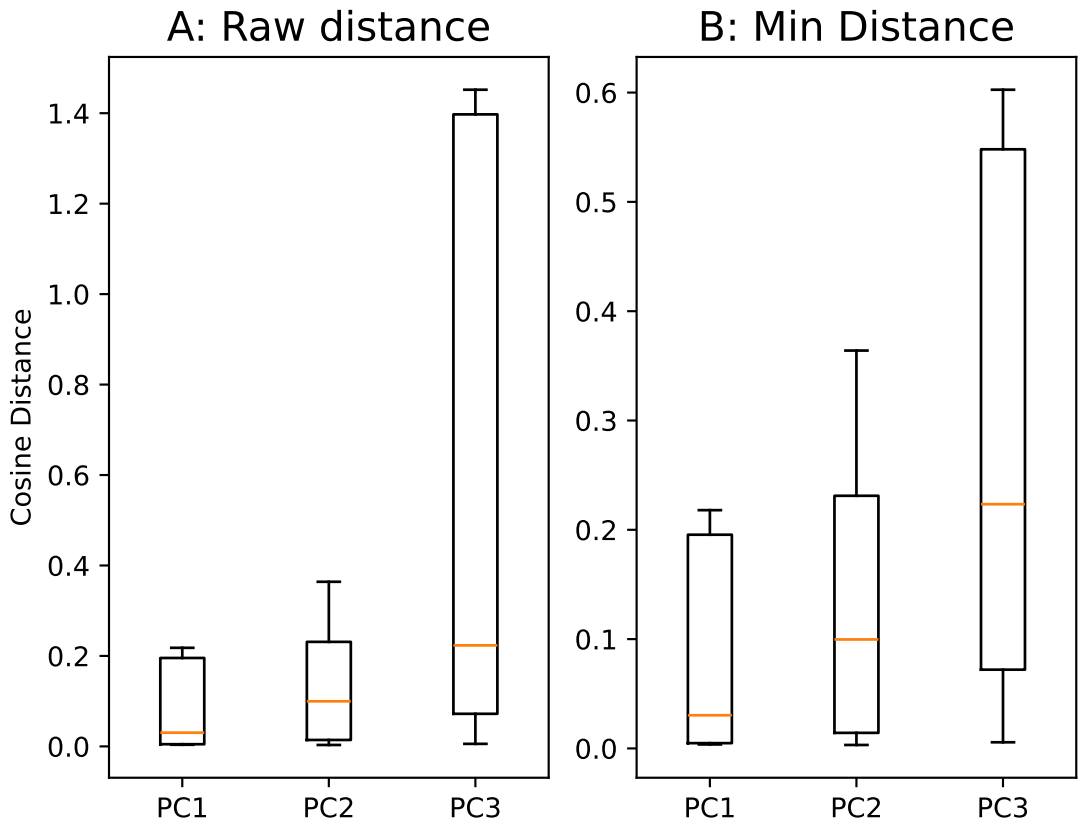

Figure 2: Distributions of distances between three first principal components. (A), raw cosine distance D between (v_1^Q, v_2^Q, v_3^Q) and (v_1^A, v_2^A, v_3^A). (B), the same as (A), but D' is displayed.

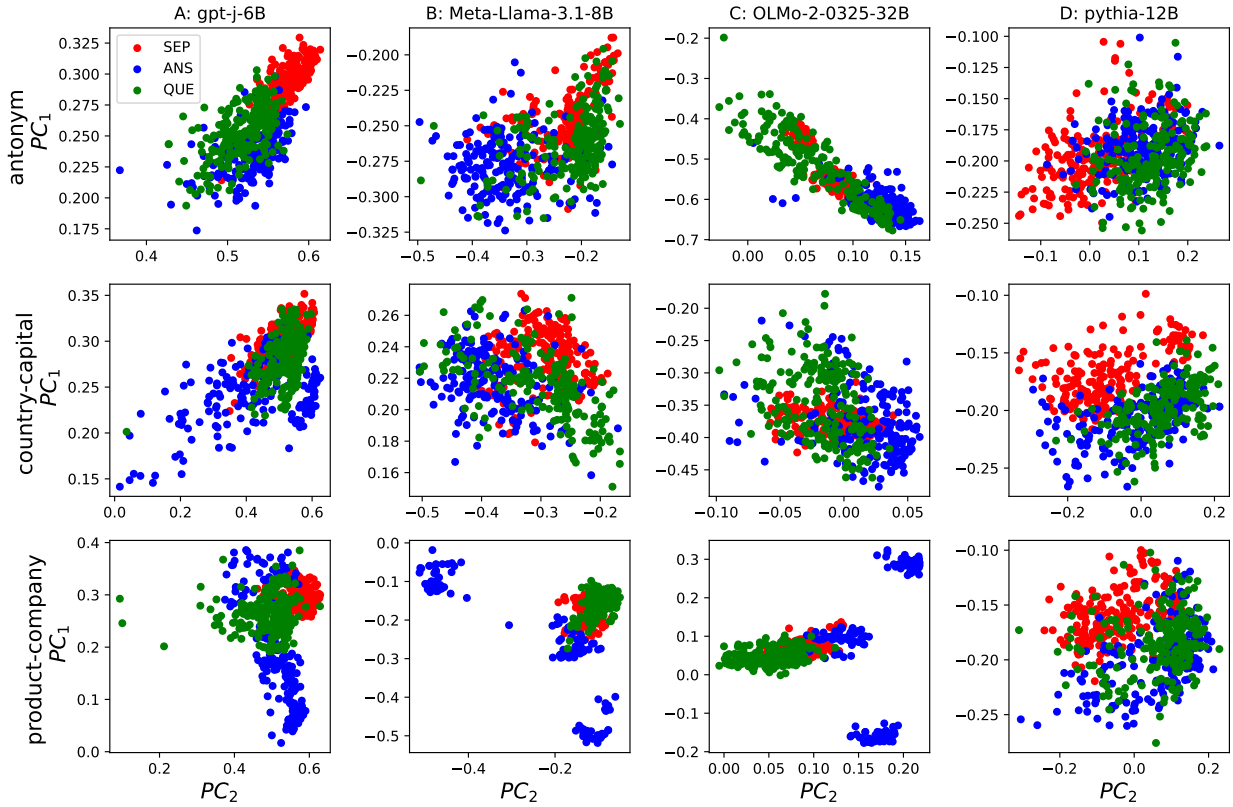

Figure 3: Tokens in the subspace spanned by v_1^A and v_2^A. We project answer (ANS), separator (SEP) and query (QUE) tokens to the subspace.(A)-(D) shows the cosine distance calcuated from gpt-j-6B, Meta-Llama-3.1-8B, OLMo-2-0325-32B and Pythia-12B. Top, middle and bottom rows denote 'antonym', 'country-capital' and 'product-company'